

PUSHKAR AWATE

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[Portfolio](#) | [LinkedIn](#) | [GitHub](#)

AI & ML Engineer • Python, AWS & Kubernetes • MLOps, Orchestration & Deployment • Agentic AI (Claude Code)

SUMMARY

AI and Machine Learning Engineer with a Master of Artificial Intelligence and two years of production infrastructure experience. I build and ship end-to-end AI and ML systems: training and tuning models, orchestrating version-controlled pipelines, and deploying and running them on cloud and Kubernetes. My most recent work is an XGBoost geospatial classification pipeline built inside an agentic, AI-agent-driven workflow.

SKILLS

Languages	Python, Java, C++, SQL, NoSQL (DynamoDB)
Cloud & Infra	AWS (EC2, ECS Fargate, Lambda, S3, SageMaker), Flask, Docker, Kubernetes (AKS, GKE), CI/CD, Git, ServiceNow
AI & Orchestration	AI coding agent (Claude Code), agentic workflows, pipeline orchestration (Kestra), Optuna, hyperparameter tuning
Geospatial & Stats	GeoPandas, QGIS, LOCO cross-validation
Machine Learning	XGBoost, TensorFlow, Scikit-learn
Monitoring & Evaluation	Model drift / OOD detection, macro-F1, error analysis, observability (Dynatrace), LLM-as-judge and agent regression gating

EXPERIENCE

Prodegee (via RMIT), Melbourne, Australia <i>AI/ML Engineering Intern</i>	<i>Feb 2026 - Jun 2026</i>
- Lifted model F1 from 0.8722 to 0.8858 across an 8-country geospatial corpus by building an XGBoost classifier and running 100 Optuna hyperparameter-tuning trials. - Selected the strongest production model by benchmarking XGBoost, Random Forest and Logistic Regression across 200 surveyed villages, then testing generalization on 35 villages with no survey data. - Restored a key feature-target correlation from -0.714 to +0.600 by tracking down and fixing a feature-collapse bug that was quietly degrading model quality. - Enabled trusted blind inference on 2,996 unsurveyed villages by building an out-of-distribution flagging system that gated low-confidence predictions. - Delivered production-ready JSON artifacts, validated in QGIS and consumed by two downstream teams, by running every task through a peer-reviewed GitHub kanban workflow with Kestra orchestration (AI coding agent: Claude Code).	
Atos-Syntel <i>Associate Consultant (Kubernetes Architect)</i>	<i>Sep 2021 - Jun 2023</i>
- Kept production Kubernetes clusters current and stable for a US-based client, as part of a seven-engineer team, by planning and running regular upgrade cycles without breaking running workloads. - Sustained high availability on critical services (Pods, Ingress) by troubleshooting incidents under time pressure, digging through logs and cluster events to find root causes. - Caught problems before they became incidents by continuously monitoring cluster health in Dynatrace, and managed incident and change workflows in ServiceNow. - Optimized cost and headroom by scaling node pools to workload demand and renewing TLS certificates on schedule. - Built deep working knowledge of core Kubernetes internals (Ingress, kube-scheduler, kube-controller-manager, kube-proxy), which sped up root-cause diagnosis when services failed.	

PROJECTS

- AI Job-Fit Agent + Live Job Board** AI agent that scores resume-to-job-description fit, flags skill gaps and drafts tailored bullets on a reusable agent runtime with a pluggable deterministic or LLM brain, then extended into a live board that pulls real job postings and ranks them by fit. Deployed on Streamlit with GitHub Actions CI. | jobfitagent.streamlit.app | github.com/pushkar-awate/jobfit-agent
- Self-Healing MLOps Pipeline** Autonomous controller that detects data drift, retrains, and promotes a new model only when it beats the current one on held-out data, with rollback and an audit ledger; runs on the real ELEC2 concept-drift benchmark and shares the same from-scratch runtime as the job-fit agent. Deployed on Streamlit with GitHub Actions CI. | selfheal-mlops.streamlit.app | github.com/pushkar-awate/selfheal-mlops
- Agent Eval & Regression Gate** Eval and regression harness for LLM agents: stores a good run as a baseline and blocks any prompt or model change that makes the agent worse. Scores each run with an LLM-as-judge (faithfulness and relevance) and a grounded-strengths guardrail, gated in GitHub Actions CI. Evaluates the real job-fit agent on live job postings and shares the same from-scratch runtime as the two projects above. Deployed on Streamlit. | agenteval-pushkar.streamlit.app | github.com/pushkar-awate/agenteval
- Music Subscription App** Deployed one Flask music service three ways (EC2, ECS Fargate and Lambda), each backed by DynamoDB and S3 and containerized with Docker, to compare cost, latency and operational overhead across VM, managed-container and serverless approaches | github.com/pushkar-awate/music-subscription-app
- Healthcare Management System** JavaFX MVC desktop application with JPA/Hibernate persistence and JUnit test coverage, built to demonstrate clean software-engineering fundamentals outside an ML context | github.com/pushkar-awate/healthcare-javaafx-assignment
- Interactive Mirror (Voice Assistance)** Voice-controlled smart mirror on Raspberry Pi combining a two-way mirror, a display and voice-assistant integration; published academically in IRJET Vol. 8, Issue 4 (Sr. No. 330) | irjet.net/volume8-issue04

EDUCATION

RMIT University, Melbourne, Australia

2024 - 2026

Master of Artificial Intelligence

Relevant coursework: Computational ML, Deep Learning, Reinforcement Learning, Cloud Computing, Intelligent Decision Making

Pune University, India

2018 - 2021

Bachelor of Information Technology

CGPA: 7.78 / 10